

Shell Sort!

Preview for Shellsort

Idea: Move entries more than one position at a time by h-sorting the array

an h-sorting array is h interleaved sorting subsequence

$h = 4$

L E E A M H L E P S O L T S X R

L M P T

E H S S

E L O X

A E L R

Preview Shellsort

Shellsort (Shell: 1956) h-sort array for decrease sequence of values of h

0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5

input - S H E L L S O R T E X A M P L E

13-sort P H E L L S O R T E X A M S L E

4-sort L E E A M H L E P S O L T S X R

1-sort A E E E H L L L M O P R S S T X

h-sorting

How to h-sort an array? Insertion sort, with stride length h

3-sorting an array

M	O	L	E	E	X	A	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T
E	O	L	M	E	X	A	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T
E	E	L	M	O	X	A	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T
E	E	L	M	O	X	A	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T
A	E	L	E	O	X	M	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T
A	E	L	E	O	X	M	S	P	R	T	A	E	L	E	O	P	M	S	X	R	T

h-sorting

- Why insertion sort?
- Big increments -> small subarray
- Small increments -> nearly in order (stay tuned)

S O R T E X A M P L E

7-sort:

S O R T E X A M P L E
M O R T E X A **S** P L E
M O R T E X A S **P** L E
M O **L** T E X A S P **R** E
M O L **E** E X A S P R **T**

3-sort:

M O L E E X A S P R T
E O L **M** E X A S P R T
E **E** L M **O** X A S P R T
E E L M O **X** A S P R T
A E L **E** O X **M** S P R T
A E L E O X M **S** P R T
A E L E O **P** M S **X** R T
A E L E O P M S X **R** T
A E L E O P M S X R **T**

1-sort

A	E	L	E	O	P	M	S	X	R	T
A	E	L	E	O	P	M	S	X	R	T
A	E	L	E	O	P	M	S	X	R	T
A	E	E	L	O	P	M	S	X	R	T
A	E	E	L	O	P	M	S	X	R	T
A	E	E	L	O	P	M	S	X	R	T
A	E	E	L	M	O	P	S	X	R	T
A	E	E	L	M	O	P	S	X	R	T
A	E	E	L	M	O	P	R	S	X	T
A	E	E	L	M	O	P	R	S	T	X

result - A E E L M O P R S T X

Shellsort: intuition

Proposition: a g-sorted array remains g-sorted after h-sorting it.

7-sort:

S	O	R	T	E	X	A	M	P	L	E
M	O	R	T	E	X	A	S	P	L	E
M	O	R	T	E	X	A	S	P	L	E
M	O	L	T	E	X	A	S	P	R	E
M	O	L	E	E	X	A	S	P	R	T

3-sort:

M	O	L	E	E	X	A	S	P	R	T
E	O	L	M	E	X	A	S	P	R	T
E	E	L	M	O	X	A	S	P	R	T
E	E	L	M	O	X	A	S	P	R	T
A	E	L	E	O	X	M	S	P	R	T
A	E	L	E	O	X	M	S	P	R	T
A	E	L	E	O	P	M	S	X	R	T
A	E	L	E	O	P	M	S	X	R	T

Still 7-sorted

Shellsort – which increment sequence to use?

Power of two: 1, 2, 4, 8, 16, 32, ...

No

Power of two minus one: 1, 3, 7, 15, 31, 63 - may be

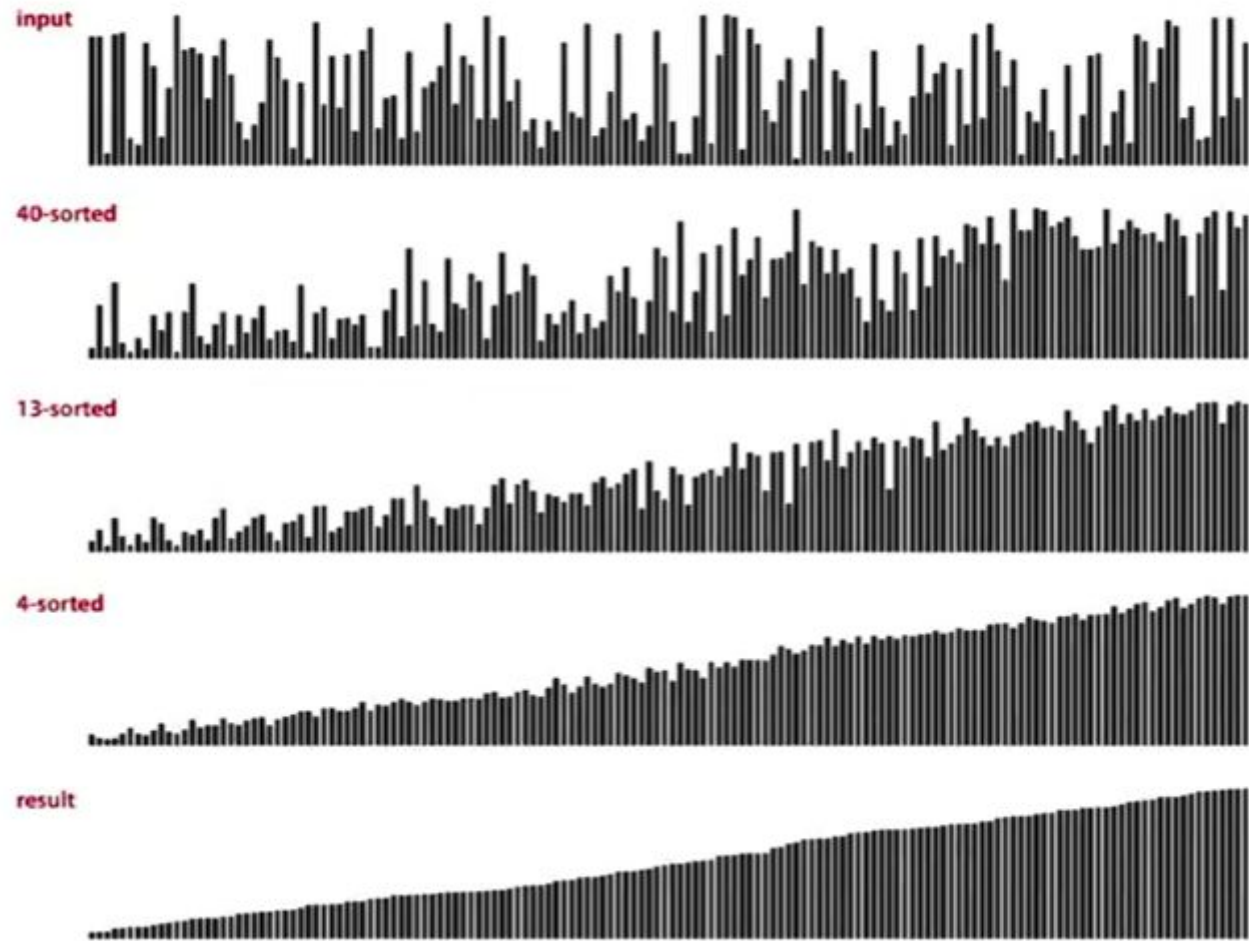
$3x + 1$: **1, 4, 13, 40, 121, 364, ...**

Sedgewick: 1, 5, 19, 41, 109, 209, 505, 929, 2161, 3905

Good – Tough to beat in empirical studies

Merging of $(9 \cdot 4^i) - (9 \cdot 2^i) + 1$
and $4^i - (3 \cdot 2^i) + 1$

Shellsort: visual trace



demo

Shellsort analysis

Proposition: The worst case number of compare used by shellsort with the $3x+1$ increments is $O(N^{3/2})$

Property: Number of compare used by shellsort with the $3x+1$ increments is at most by a small multiple of N times the # of increments used

Why are we interested in shellsort?

Example of simple idea leading to substantial performance gains

Useful in practice:

- Fast unless array size is huge
- Tiny, fixed footprint for code (used in embedded systems)
- Hardware sort prototype

Simple algorithm, nontrivial performance, interesting questions:

- Asymptotic growth rate
- Best sequence of increments <- open problem: find a better increment sq
- Average-case performance

QUIZ

How many compares does shellsort (using the $3x+1$ increment sequence) make on an input array that is already sorted?

1. Constant
2. Logarithmic
3. Linear
4. linearithmic

QUIZ

How many compares does shellsort (using the $3x+1$ increment sequence) make on an input array that is already sorted?

1. Constant
2. Logarithmic
3. Linear
4. linearithmic

Since successive increment values of h differ by at least a factor of 3, there are $\sim \log_3 n$ increment values. For each increment value h , the array is already h -sorted so it will make $\sim n$ compares.

s = S O R T E X A M P L E

4-sort

E O R T S X A M P L E
E O R T S X A M P L E
E O A T S X R M P L E
E O A M S X R T P L E
E O A M P X R T S L E
E L A M P O R T S X E
E L A M P O E T S X R

1-sort

E L A M P O E T S X R
A E L M P O E T S X R
A E L M P O E T S X R
A E L M P O E T S X R
A E E L M O P T S X R
A E E L M O P T S X R
A E E L M O P S T X R
A E E L M O P S T X R
A E E L M O P R S T X

SORTEXAMPLE

4-sort

EORT**S**XAMPLE
EORTS**X**AMPLE
EO**A**TSX**R**MPLE
EOA**M**SXR**T**PLE
EOAM**P**XRT**S**LE
E**L**AMP**O**RTS**X**E
ELAMP**O**ET**S**X**R**

ELAMPOETSR

1-sort

E**L**AMPOETSR
AELMPOETSR
AEL**M**POETSR
AELM**P**OETSR
AELM**OP**ETSR
AE**ELMOP**TSR
AEELMOP**TS**XR
AEELMOP**ST**XR
AEELMOP**RSTX**

Final Result : A E E L M O P R S T X